

AI-Based FDIA Detection in IIOT

Dual-Head GNN with Adaptive RL Framework for FDIA on IIOT

Introduction:

We introduce ASTRO, an Adaptive Spatio Temporal Reinforcement Optimization framework that detects stealthy False Data Injection Attacks across the Industrial Internet of Things in real time.



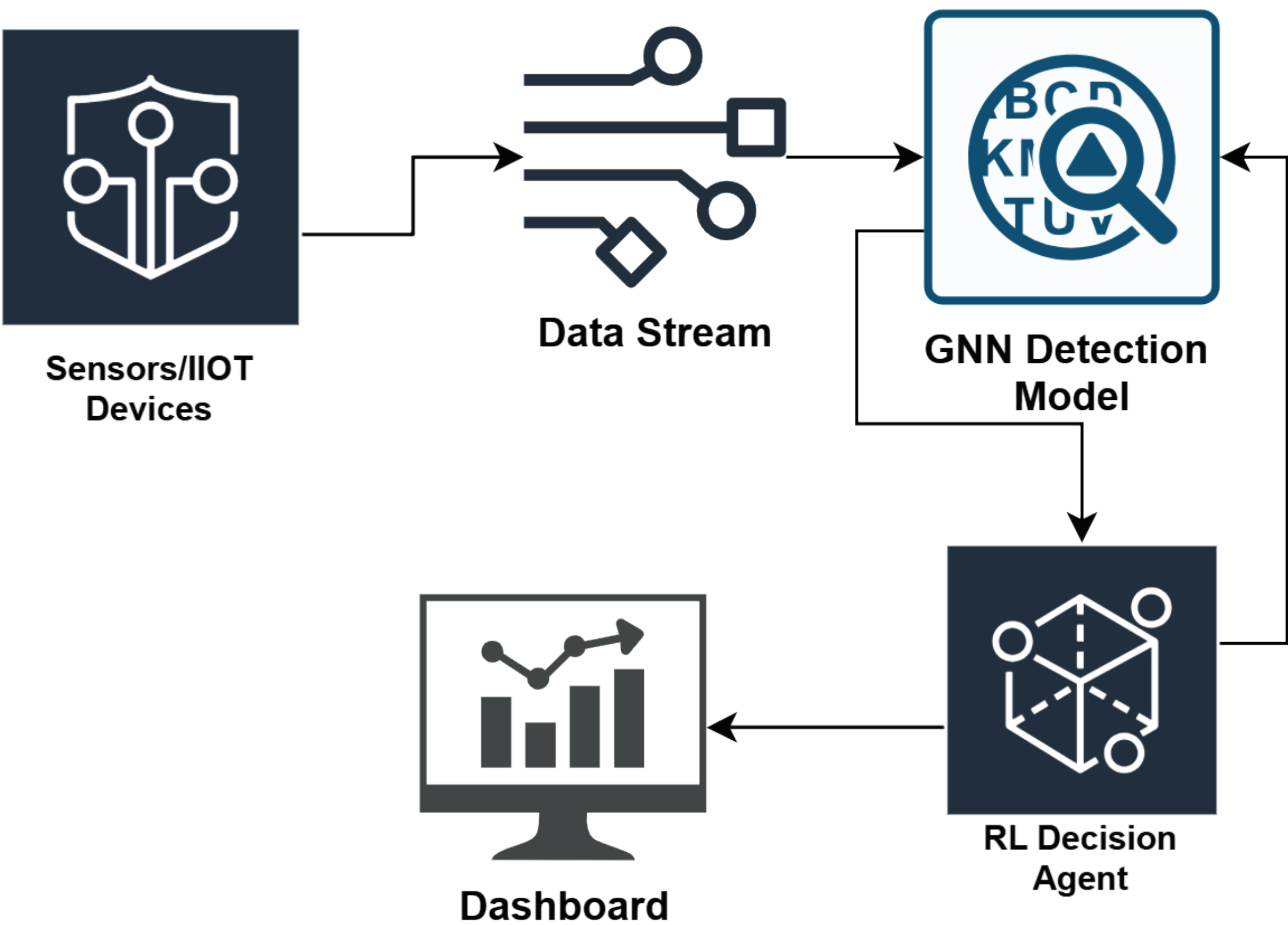
Objectives:

- Detect stealthy FDIAs in IIoT networks.
- Map sensor topologies using Dual-Head GNNs.
- Automate intelligent responses with Adaptive RL.
- Safeguard critical industrial operations.

Methodology:

False Data Injection Attacks threaten the safety of the Industrial Internet of Things by secretly manipulating sensor data. We introduce ASTRO, an Adaptive Spatio Temporal Reinforcement Optimization framework designed to detect these stealthy threats in real time. It achieves this by mapping the physical network through a novel graph of inter machine and intra machine connections.

Architecture Diagram:



Results:

Performance Metrics on Test Data

Metric	Value
Accuracy	0.9902
Precision	0.9831
Recall	0.9975
F1-Score	0.9902

Timeline:



Tools & Technologies:



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